Diagnostic Report generated by gemini-2.5-pro

Diagnostic Analysis of Run_35

1. Summary of Key Deviations

The analysis of Run_35 reveals a significant physical fault characterized by a system-wide loss of pressure and flow, with a particularly unusual signature in the bypass loop.

- **p_Combustion (Moderate):** The most significant and consistent anomaly is a major **negative offset** on the combustion pressure, which reads approximately **-24%** lower than healthy across all active states.
- **Flow_Bypass (Moderate):** The bypass flow shows a large **negative offset** of approximately **-26%**. This is a key diagnostic clue.
- **Flow_Pump (Moderate):** The pump flow shows a corresponding moderate **negative offset** of approximately **-12%**.
- **Flow_PRV (Moderate):** Uniquely, the `Flow_PRV` signal shows a moderate positive **offset** of **+13.8%** at the very beginning of the run (state s1), indicating the high-pressure relief valve is active even at idle.
- **Other Signals (Nominal/Minor):** All other signals are either nominal or show only minor, likely secondary, deviations.

2. Root-Cause Hypotheses

Rank	Suspected Component	Fault Type	Supporting Evidence
1	**Pressure Relief Valve (PRV)**	**Stuck Partially Open / Low Cracking Pressure**	This is the most compelling hypothesis. The fact that the **PRV is flowing at idle** (state s1) is highly anomalous and indicates it's opening at a much lower pressure than it should. A leaking PRV creates a permanent loss of pressure and flow from the main system, perfectly explaining the major negative offset on **p_Combustion** (-24%), as well as the negative offsets on **Flow_Pump** (-12%) and **Flow_Bypass** (-26%). The system is constantly losing a portion of its output through the incorrectly opened PRV.

2	**Combustion Chamber / Injector Nozzle(s)**	**Major Leak**	A major leak in the combustion chamber would explain the -24% pressure drop. However, it doesn't explain why the `Flow_PRV` was active at idle. A leaking PRV is a more direct cause that explains all symptoms, including the PRV flow itself.
3	**Main Fuel Pump**	**Moderate Degradation**	A degraded pump would explain the drop in `Flow_Pump` and the corresponding drop in pressures. However, it fails to explain the anomalous PRV activity at the start of the run.
4	**Bypass Valve**	**Stuck Significantly Open**	A stuck-open bypass valve would cause `Flow_Pump` and `Flow_Bypass` to be low. However, it typically wouldn't cause such a large drop in `p_Combustion` and wouldn't explain the PRV flow at idle. The PRV leak is a better fit for the specific evidence.
5	**Multiple Sensor Failures**	**Various Negative Offsets**	It is highly improbable that `p_Combustion`, `Flow_Pump`, and `Flow_Bypass` sensors would all fail simultaneously with a negative offset. The physically consistent nature of the deviations points strongly to a single, real physical fault.

3. Most Probable Hypothesis

The most probable root cause is that the **Pressure Relief Valve (PRV) is Stuck Partially Open or has a low cracking pressure.**

Reasoning: The "smoking gun" for this diagnosis is the `Flow_PRV` reading of +13.8% in state s1 (idle). The PRV should only ever activate during a severe over-pressure event, never at idle. This indicates the valve is faulty and is leaking fuel out of the main circuit from the very beginning of the run. This single, continuous leak provides a perfect explanation for all other symptoms: the system is constantly losing pressure and flow, causing `p_Combustion` to be low, and forcing both `Flow_Pump` and `Flow_Bypass` to operate at a reduced level.